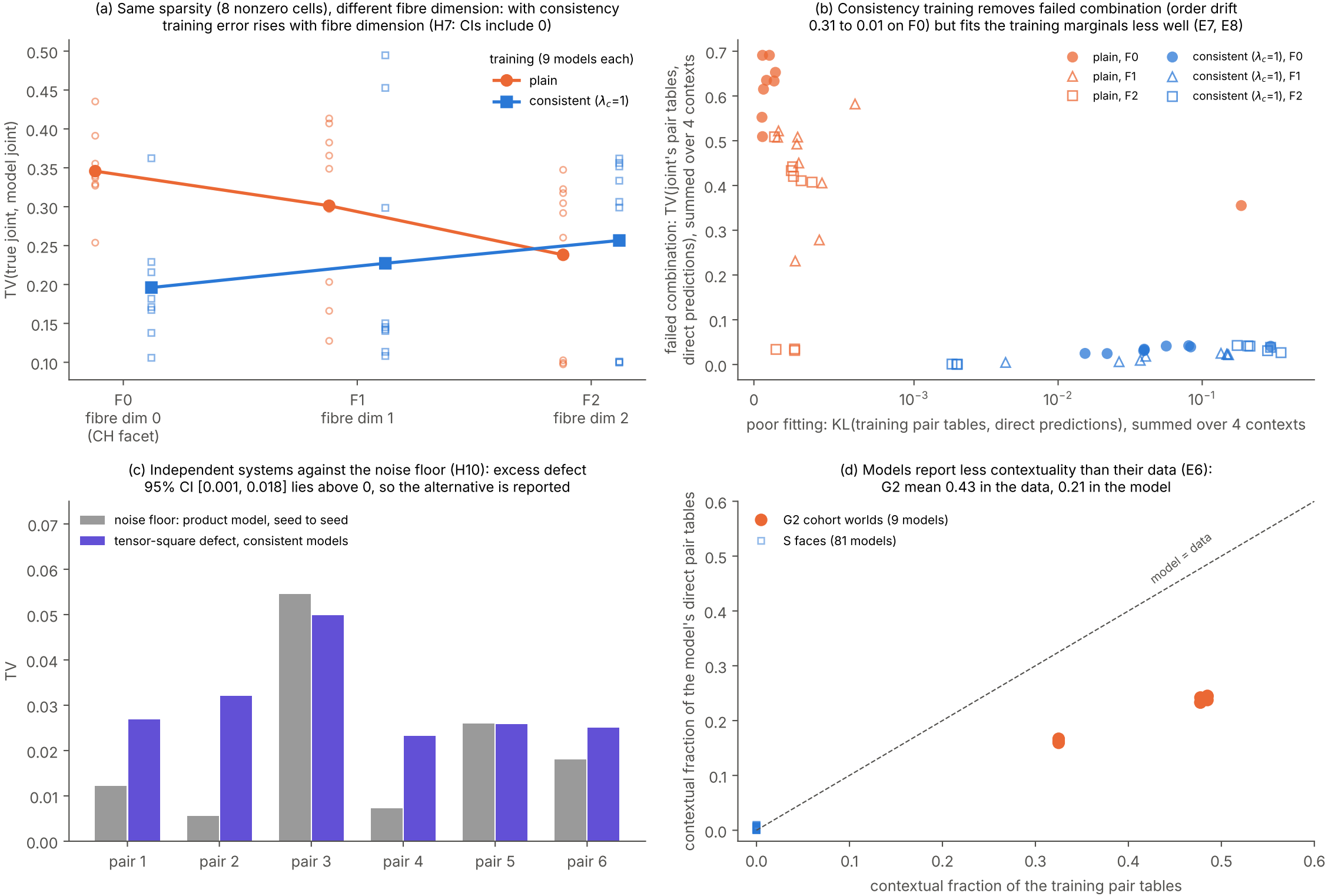
(a) Same sparsity (8 nonzero cells), different fibre dimension: with consistency training error rises with fibre dimension (H7: CIs include 0)
training (9 models each)
plain
consistent ($\lambda_c$=1)
TV(true joint, model joint)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
F0
fibre dim 0
(CH facet)
F1
fibre dim 1
F2
fibre dim 2
(b) Consistency training removes failed combination (order drift 0.31 to 0.01 on F0) but fits the training marginals less well (E7, E8)
plain, F0
plain, F1
plain, F2
consistent ($\lambda_c$=1), F0
consistent ($\lambda_c$=1), F1
consistent ($\lambda_c$=1), F2
failed combination: TV(joint's pair tables, direct predictions), summed over 4 contexts
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
$10^{-3}$
$10^{-2}$
$10^{-1}$
poor fitting: KL(training pair tables, direct predictions), summed over 4 contexts
(c) Independent systems against the noise floor (H10): excess defect 95% CI [0.001, 0.018] lies above 0, so the alternative is reported
noise floor: product model, seed to seed
tensor-square defect, consistent models
TV
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0.00
pair 1
pair 2
pair 3
pair 4
pair 5
pair 6
(d) Models report less contextuality than their data (E6): G2 mean 0.43 in the data, 0.21 in the model
G2 cohort worlds (9 models)
S faces (81 models)
model = data
contextual fraction of the model's direct pair tables
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
contextual fraction of the training pair tables